

Lab 13: Shrinkage Estimation

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13.1 A Putting Prediction Contest

Use `13_putts-train.csv` to predict each golfer's held-out putting rate in `13_putts-test.csv`. The test outcomes may be used only for final evaluation.

Your primary score is mean squared error across all golfers:

$$\text{MSE} = \frac{1}{k} \sum_{i=1}^k (\hat{p}_i - X_i^{\text{test}})^2.$$

Fit four estimators:

$$\begin{aligned}\hat{p}_i^{\text{mean}} &= \bar{X}, \\ \hat{p}_i^{\text{MLE}} &= X_i, \\ \hat{p}_i^{\text{EB}} &= \hat{\mu} + \frac{\hat{\tau}^2}{\hat{\tau}^2 + \hat{C}/N_i} (X_i - \hat{\mu}), \\ \hat{p}_i^{\text{JS}} &= \bar{X} + \left[1 - \frac{(k-3)\hat{\sigma}^2}{\sum_j (X_j - \bar{X})^2} \right]_+ (X_i - \bar{X}),\end{aligned}$$

where $\hat{\sigma}^2 = \text{Mean}(\hat{C}/N_i)$ is a common-variance approximation for the James–Stein calculation. Because this rule estimates its center using $\bar{X}$, its dominance result requires at least four coordinates.

13.1.1 Task 1: Fit Without Looking at the Test Set

Using only the training data, calculate all four estimators. Keep predictions inside $[0, 1]$. Describe the shrinkage target and strength used by each estimator.

Explain what is lost by replacing the player-specific sampling variances $\hat{C}/N_i$ with one common variance. Describe how standardizing deviations from the league mean could instead create approximate common-variance observations.

13.1.2 Task 2: Compare the Leaderboards

For each method, rank the golfers from best to worst. Compare:

- the range and standard deviation of predictions;
- the identities of the top and bottom five golfers;

- how far each prediction moves from the MLE.

Explain why the raw leaderboard is expected to be more dispersed.

13.1.3 Task 3: Reveal the Test Set

Join the test outcomes and compute MSE across golfers for every method. Make a plot comparing predictions with test performance and rank the estimators by MSE.

13.1.4 Task 4: The Consultant's Dilemma

For every golfer, calculate the squared error of the MLE and your best shrinkage estimator. Make a plot showing which estimator wins for each golfer.

Does the method with the best overall MSE win for every individual? Explain why this does or does not contradict the lecture's shrinkage result.

13.1.5 Task 5: Change the Loss Function

Suppose overrating a golfer is twice as costly as underrating one. Define and compute an asymmetric loss for all four estimators. Does your preferred estimator change? Explain why the loss function is part of the modeling decision.